LingAfriq: Premium Curriculum Plan (accurate before native authors)

Prepared for: Tope (Fabian Umole) · **Date:** 17 October 2026 · **Companion to:** `LINGAFRIQ_ELEVATION_PLAN.md` (§6 curriculum, §17–18 features) and `LINGAFRIQ_AI_MODEL_STACK.md` (models, costs).

The brief. Build highly accurate, premium curricula for the Stage-1 seven (Yoruba, Igbo, Hausa, Nigerian Pidgin, Swahili, isiXhosa, isiZulu) and then as many African languages as possible, **without native speakers writing them.** Native speakers come later as final validators. Until then, the content must be as accurate as it can be, and the app must never present an unverified item as certain.

Method. Licences and availability were checked on the live web on 17 Oct 2026 (links inline). Legal points are a product manager's reading, not legal advice; get counsel to sign off the licence register (§2.4). Costs and accuracy figures are planning estimates, marked as such, to be replaced by measured numbers from the pilot.

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0. The plan on one page

1. **Ground every item in a source, never in the model's memory.** Lexemes, tone patterns, grammar rules and culture claims come from a licence-checked **source library** (public-domain FSI courses, commercially licensed dictionaries, CC-BY corpora, official orthography rules). An LLM drafts only **inside** that evidence and must cite it claim by claim.
2. **Verify with machines that disagree with each other.** A second model family cross-checks, independent MT back-translates, deterministic linters check orthography, tone marks and Bantu concords, and morphological analysers parse Zulu. Every item gets a **confidence score**: high scores publish as Beta, mid scores go to a human queue, low scores are rejected and regenerated.
3. **Buy expert spot-checks instead of authors.** Paid linguists and academics review **100% of high-risk items** (tone marks with no lexicon match, idioms, culture and history, honorifics, grammar explanations) and a **statistical sample** of the rest. That costs a fraction of authoring.
4. **Label honestly.** Items show **Beta**, **Expert-reviewed** or **Native-verified**; synthetic audio says "Synthetic voice". Learners can tap "Report a mistake" on every item, and errors flow back into the pipeline.
5. **Rebuild today's broken data through the same pipeline.** Pull placeholders from the build, collapse the three identical A1–C2 copies into one signed content pack, replace the two copy-pasted grammar lines with a cited Pattern Library, and correct false tone claims by lexicon lookup.
6. **Realistic outcome:** Beta A1 for all seven languages in about **12 weeks**, A2 in about 24 weeks, and B1 Expert-reviewed by about month 9 (all in Beta first). Per language, roughly **$4k–15k** in content cost to reach A1–B1 Expert-reviewed, plus one-off licences and tooling (§9). Accuracy targets: §6.

Step	1	2	3	4	5	6
Flow	Source library (licence-checked)	Grounded LLM draft, cited per claim	2nd model + back-translation + linters + analysers	Confidence score	**High:** Beta + expert sample · **Mid or high-risk:** linguist → Expert-reviewed · **Low:** reject and regenerate	Learner flags and item stats feed back; native validators later → Native-verified

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1. What is wrong with today's curriculum (from the audit)

Problem (evidence in elevation plan §6.1)	Scale	How the pipeline fixes it (details in §8)
Placeholder bundles (`"word1a"`, `"example meaning"`, `"Sample MCQ"`, answer `"a"`)	**24,234 of 48,847** vocabulary entries (49.6%)	CI placeholder linter blocks the build; bundles leave the app; units are rebuilt from the source library
"Authentic" A1–C2 file is thin	85–111 unique words per language across 126 lessons; 126/126 dialogues repeat lines	Real phrases are salvaged as seeds and re-verified; the frequency-based lexicon targets 2,500–3,500 lexemes for A1–B1
Triple-copied A1–C2 file	3 × 11.3 MB identical (same md5), about 34 MB of APK	One source of truth; versioned, signed content packs downloaded per language and level
The same two grammar sentences pasted into 1,546 lessons, false for many languages ("affixes carry tense and person" is wrong for Yoruba and Pidgin, which use preverbal particles)	1,546 lessons	A per-language **Pattern Library** written from cited reference grammars; a claim linter rejects generic grammar text
False tone teaching ("High tone on àárọ̀ marks respect"; `phonetic_guide` copies `tonal_note` for all 1,409 words)	1,409 game words	The tone-claim linter requires a lexicon source for every tone statement; IPA is generated by a deterministic grapheme-to-phoneme converter (G2P)
UI and phrase packs without diacritics ("E kaabo", "Omo")	`app_yo.arb` 62/103 keys; offline packs	The same orthography linters run on UI strings and phrase packs
Audio filenames in non-ASCII orthography; zero audio references in the curriculum	1,363 manifest entries	Content-hash filenames; every item carries an audio record (TTS-interim or human)

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2. Source-grounded authoring: the source library

2.1 Rules of use

- **Facts versus expression.** That *ọkọ̀* means "vehicle" with a mid–low tone pattern is a linguistic fact; a dictionary's wording of the definition and its example sentences are the publisher's expression. Learner-facing text is **written fresh by the pipeline** and cites sources as evidence. Copy verbatim only from public-domain or compatible-licence sources, with attribution. (EU database right can also protect substantial extraction from a database; counsel should confirm per source.)
- **Share-alike containment.** CC BY-SA material (Wiktionary, UD treebanks, Biblica open Bibles, most WAXAL subsets) is used **as verification evidence**, or kept in a clearly separated, attributed CC BY-SA layer. It is **not** mixed into LingAfriq's proprietary lexicon text, because that would put the derived database under CC BY-SA.
- **Non-commercial means not usable** in a paid app without a written licence or waiver. Using such data only to train or check models is still commercial use of the data; treat it as excluded unless licensed.
- Every source gets a **licence register** entry: URL, licence, version/date, allowed uses (display, evidence, training), attribution text, and owner sign-off.

2.2 General sources (all languages)

Source	Licence (verified 17 Oct 2026)	Commercial use?	How to use it
FSI Basic Courses (Yoruba 380 pp + tone drills + Intermediate Texts; Igbo 1962, 30 units; Hausa; Swahili 1968, 150 units). Yoruba, Igbo, Swahili, archive	US government work, public domain	**Yes**	Grammar sequencing, tone drills, dialogues as evidence; old audio as tone references. Language is 1960s, so modernise vocabulary. No FSI Zulu or Xhosa course found
Peace Corps manuals (Hausa, Kiswahili, Zulu; Yoruba 1965 on ERIC)	Often US-gov public domain, but **not established for every item** (contractor-written works may not be). Live Lingua says to contact Peace Corps for commercial use	**Verify per item**	Can-do topics, survival dialogues, culture notes as evidence; confirm status before copying
Wiktionary / Kaikki.org (licence, Kaikki)	CC BY-SA 4.0 + GFDL; wiktextract code is MIT	Yes, with attribution **and share-alike**	Cross-check evidence for lemma, POS, tone marks, inflection; do not copy into the proprietary lexicon (§2.1)
Leipzig Corpora Collection downloads (terms)	Downloadable corpora CC BY; web services and applications CC BY-NC	**Yes** (downloads, with attribution)	**Frequency lists** and example-sentence evidence per language [verify each language's corpus size]
Mozilla Common Voice (Yoruba 26.0, Hausa, Igbo, Swahili 27.0)	CC0	**Yes**	Read-speech audio for listening examples and TTS/ASR evaluation. Recorded hours: Yoruba 8.3, Hausa 12.6, Igbo 18.9, Swahili 1,065 [Zulu and Xhosa hours: verify]
Google FLEURS (card)	CC BY 4.0	**Yes**	Parallel sentences plus audio for yo, ig, ha, sw, xh, zu: an MT and ASR evaluation set
Google WAXAL (2026; dataset, blog)	**Mixed:** University of Ghana subsets CC BY 4.0, most others CC BY-SA 4.0	Yes, per subset	Speech for evaluation and TTS research; check each subset's licence
African Storybook (terms, source repo)	Per story: CC BY **or** CC BY-NC	CC BY stories only	**Graded readers** (zu, xh, sw, yo, ha). Filter to CC BY; attribute authors; check for Igbo
Universal Dependencies (Yoruba-YTB, Naija-NSC)	CC BY-SA 4.0	Yes, share-alike	Parsed sentences as grammar evidence (Yoruba, Pidgin)
Biblica open Bibles on eBible (Yoruba, Hausa)	CC BY-SA 4.0 (Biblica trademark removed from derivatives)	Yes, share-alike	Corpus and frequency evidence only; religious register. Other Bible translations (e.g., Bible Society editions) are copyrighted
Lanfrica catalogue (about)	Catalogue of links with licence metadata	n/a	Discovery only; always verify the licence at the source
Licence-audit paper (arXiv 2606.28867)	Research	n/a	Use its compatibility matrix as a checklist for the register

2.3 Excluded or "needs a licence"

Source	Status	What to do
JW300	Built in breach of jw.org terms, which prohibit text and data mining; OPUS removed it (CIPIT/infojustice, OPUS issue)	**Exclude.** Declare "no JW300-derived data" in the register

Source	Status	What to do
MAFAND-MT (README), **MasakhaNER 2.0** (repo)	CC BY-NC 4.0	Exclude, or request a commercial licence from Masakhane
AfriSenti (HF, paper)	HF card says CC BY 4.0, but the paper restricts commercial use without creator approval	Treat as non-commercial; ask the creators. Low value for curriculum anyway (tweets)
PanLex (licence)	CC BY-NC-SA 4.0; commercial use needs written permission (some HF derivatives still claim CC0 from an older licence)	Request permission (info@panlex.org) before any use
NaijaVoices (HF, licence, membership)	CC BY-NC-SA 4.0; commercial waiver via membership ($5,000+ tier, 12 months, revocable)	Buy the waiver if you need Igbo/Hausa/Yoruba speech at scale
IgboAPI dataset (paper)	Data CC BY-NC-SA 4.0 (code Apache-2.0)	High value (multi-dialect Igbo with examples): **license it** from Nkọwa okwu
Kamusi (policy)	CC BY-NC-SA; commercial licence on request	License it if a Swahili gap remains after Oxford
Helsinki Corpus of Swahili 2.0 (Kielipankki, ELRA)	CLARIN ACA (academic only); commercial licence via ELRA €7,500 (member) / €15,000	Optional; Leipzig CC BY plus Common Voice may suffice for frequency
Oxford Languages datasets (datasets, API languages)	Commercial licence; lists Hausa, isiXhosa, Igbo, Yoruba datasets, with Swahili via the API	**Buy (top pick for accuracy).** Get a quote covering tone-marked headwords, examples and the right to display
Classic dictionaries (Abraham, *Dictionary of Modern Yoruba*, 1958, archive.org; Bargery, *Hausa–English*, 1934, online)	Copyright status unconfirmed (author-death terms likely still running in the UK)	Reference and verification only; counsel to confirm

2.4 Per-language source stack (Stage 1)

Language	Orthography authority	Lexicon (primary → check)	Grammar evidence	Corpora / audio	Tools
Yoruba	Standard Yoruba orthography (Nigerian standard; full tone marking for pedagogy) [cite NERDC edition: verify]	Oxford Yoruba dataset (licensed) → Wiktionary, Abraham (reference)	FSI Basic + Intermediate (PD); a modern reference grammar, e.g. Bámgbóṣé *A Grammar of Yoruba* [purchase; not web-verified]; UD-YTB	Leipzig, Biblica open Bible, Common Voice, FLEURS	NaijaML diacritizer, Apache-2.0: 80.3% full-tone word accuracy, 93.3% dot-below (repo); Spitch tone marking (AI stack)
Igbo	Standard Igbo (Ọnwụ orthography) [verify]; dialect policy: Central Igbo plus Onitsha/Owerri notes	IgboAPI (license) + Oxford Igbo → Wiktionary	FSI Igbo Basic (PD, tone drills; tape 1 missing); Emenanjo's Igbo grammar [purchase; verify]	Leipzig, Common Voice, FLEURS	Igbo tone restorer only **61.6%** tone-mark accuracy (model), so Igbo tone is **lexicon-first, human-checked**
Hausa	Standard Boko orthography (ɓ ɗ ƙ y ’). Tone and vowel length are unwritten in normal text, so **pedagogical marking** comes from dictionaries	Oxford Hausa → Wiktionary, Bargery (reference)	FSI Hausa Basic (PD); Peace Corps Hausa; Newman *The Hausa Language* (Yale, 2000) [purchase]	Leipzig, Biblica open Bible, Common Voice, FLEURS	Lexicon lookup for tone and length marks

Language	Orthography authority	Lexicon (primary → check)	Grammar evidence	Corpora / audio	Tools
Nigerian Pidgin (Naijá)	**Naija Langwej Akademi standard orthography** (IFRA guide): phonetic Latin, high tone marked only in scientific use	No commercial dictionary found; build from UD-Naija, Faraclas *Nigerian Pidgin* (1996) [purchase] and corpora	UD-Naija-NSC (CC BY-SA)	NSC speech corpus [verify licence]; Intron for ASR	LLM "anglicisation" detector (§3.4)
Swahili	Kiswahili Sanifu (BAKITA, Tanzania) [verify citation] + Kenyan notes	Oxford API Swahili → Wiktionary; Kamusi (license, optional)	FSI Swahili Basic (PD, 150 units); Peace Corps Kiswahili; Ashton *Swahili Grammar* [purchase]	Leipzig, Common Voice (1,065 h), FLEURS, African Storybook	Noun-class concord checker (§3.3)
isiZulu	PanSALB revised spelling and orthography rules (Gazette, 6 Jul 2023)	Commercial Zulu dictionary licence [Oxford API doesn't list Zulu; approach the publisher of the Doke–Vilakazi dictionary / the isiZulu National Lexicography Unit: verify] → Wiktionary	Peace Corps Zulu; Poulos & Msimang [purchase]; SADiLaR morphologically annotated corpus, 45,933 tokens, **CC BY 4.0** (repo)	NCHLT corpora (CC BY 2.5 ZA), FLEURS, WAXAL, African Storybook	**ZulMorph** finite-state analyser; the SADiLaR record lists Apache-2.0 (record, demo) [verify local-use terms]
isiXhosa	PanSALB revised rules (same Gazette)	Oxford isiXhosa dataset (license) → Wiktionary	SADiLaR annotated corpus, 46,465 tokens, CC BY 4.0 (repo); a standard Xhosa reference grammar [linguist to choose]	NCHLT, FLEURS, WAXAL, African Storybook	Concord checker; no mature open Xhosa analyser confirmed [verify]

Later waves. Reuse the same template. Each new language needs: 1. an orthography authority; 2. one licensed or public-domain lexicon; 3. one reference grammar; 4. a CC-BY or PD corpus.

Use Lanfrica to discover sources, and check each one against FSI's and the Peace Corps' language lists. A language **without** a lexicon source cannot pass the gates beyond L0 "Taste" until a linguist builds a seed lexicon (§6).

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3. Multi-model verification pipeline

3.1 Stages

1. **Language kit (once per language).** Allowed character set and Unicode rules (NFC); tone-marking policy for pedagogy; phone inventory; G2P rules; noun-class and concord tables (Bantu); dialect standard; a register of honorifics.
2. **Lexicon first.** Build the frequency list from Leipzig and other CC-BY corpora, then map each frequent form to a **sourced lemma** (licensed dictionary first, Wiktionary as a cross-check). Each lexeme stores tone pattern, IPA from G2P, POS, noun class, plural and sources. A lexeme with **two agreeing sources** is "anchored"; with one source, it is "single-source" (held for linguist review if high-frequency).
3. **Grounded generation.** Gemini 3.1 Pro (AI stack §3.1/§3.9) receives the can-do target, the allowed lexicon (anchored lexemes only), the relevant Pattern Library entries and retrieved evidence passages. It drafts sentences, dialogues, exercises, distractors and explanations as **structured JSON with a citation per claim** (source id plus locator). Forbidden: any word outside the allowed lexicon (it must be requested as a new lexeme), and any culture or grammar claim without a citation.

4. **Independent cross-check.** A **different model family** (GPT-5 class; a third model as tie-breaker) receives the item **without** the generator's reasoning. It translates the item to English, identifies errors, rates naturalness and register, and checks each citation supports its claim (entailment against the retrieved passage).
5. **Back-translation.** Independent MT (Google Cloud Translation; MADLAD-400 for the long tail, AI stack §3.7) translates the target text back to English, scored by semantic similarity against the intended meaning. Disagreement flags the item; it doesn't auto-reject it, because MT itself is weak for some languages.
6. **Deterministic linters** (§3.3) and **analysers** (ZulMorph for Zulu; rule-based concord checkers elsewhere).
7. **Tone and diacritic consistency.** Every token is matched to a lexicon entry, and its tone marks must equal the lexicon's for that sense. Where a restoration model exists (NaijaML for Yoruba), it is run as an **independent opinion**: the item passes only if the lexicon form, the generator's form and the restorer agree. Any disagreement goes to a linguist.
8. **Confidence scoring and routing** (§3.4).
9. **Human review** where routed (§5), then publishing with a label.

3.2 What each stage catches

Error type	Main catcher	Backup
Wrong word or meaning	Lexicon anchoring + back-translation	Cross-check model; linguist sample
Missing or wrong tone marks (yo, ig)	Lexicon tone match + restorer agreement	Linguist (100% where unmatched)
Hausa tone and vowel length (pedagogical)	Dictionary-sourced marked forms only	Linguist
Noun-class or concord errors (sw, zu, xh)	Concord linter + ZulMorph parse	Cross-check model
Unnatural or anglicised phrasing (esp. Pidgin)	Cross-check naturalness rating + anglicisation detector	Learner flags; native validators later
False grammar or culture claims	Claim-level citation + entailment check	Linguist or cultural reviewer (100%)
Wrong register or honorifics (Yoruba ẹ/ọ, Zulu hlonipha)	Honorific register list + cross-check	Linguist (100% for respect-sensitive units)
Placeholders, empty fields, broken JSON	Schema validator + placeholder regex	CI blocks the build

3.3 Linters (deterministic, in CI and in the studio)

- **Unicode:** NFC only; allowed code points per language; no look-alike characters (e.g., Latin ɛ vs Greek ε).
- **Yoruba:** dot-below vowels and ṣ; tone policy (high ´ and low ` marked, mid unmarked; syllabic nasals carry tone); every token's diacritics equal its lexicon form; flags unmarked tokens in learner-facing text.
- **Igbo:** dotted vowels ị ọ ụ and ṅ; pedagogical tone policy fixed by the linguist in the language kit [Igbo marking conventions vary; decide once]; lexicon match.
- **Hausa:** hooked letters ɓ ɗ ƙ y and ’; pedagogical marks only from sourced forms; distinguishes display text (standard, unmarked) from teaching text (marked).
- **Swahili, Zulu, Xhosa concords:** a noun's class prefix determines the subject, object, adjectival, relative and possessive concords in the clause, checked against class tables (Swahili m/wa, ki/vi ...; Zulu and Xhosa classes 1–15). Zulu forms are additionally parsed by ZulMorph; parse failure means human review. Conjunctive orthography follows the PanSALB rules (Zulu and Xhosa write verbal complexes as one word).
- **Pidgin:** NLA spelling normalisation table; flags English spellings where an NLA form exists.

- **Claims:** any sentence in an explanation or culture note that states a rule, fact, date or meaning must carry a citation id. Generic templates ("affixes carry tense and person", "Pattern: subject → verb → object") are **banned strings**.
- **Placeholders:** `/word\d|example meaning|sample|lorem|TODO/i` and duplicate-content detection (near-identical dialogues and repeated lines).

3.4 Confidence score and routing

Score = a weighted blend, calibrated on the linguist-reviewed pilot set. Inputs: - lexicon coverage (% of tokens anchored); - tone agreement (lexicon, generator and restorer); - cross-model agreement and naturalness; - back-translation similarity; - linter results; - citation entailment; - source tier (licensed dictionary or PD course > CC-BY corpus > Wiktionary-only).

Route	Rule	Result
Auto-Beta	Score ≥ 0.90, all linters pass, no high-risk tag	Publish **Beta**; enters the linguist **sample**
Review	0.70–0.90, **or any high-risk tag** regardless of score	Linguist queue; publish as **Expert-reviewed** when approved
Reject	< 0.70 or any hard linter failure	Regenerate (max 2 attempts), then park for a human

High-risk tags (never AI-only): - tone-marked tokens without a lexicon match; - idioms and proverbs; - culture, history and religion claims; - honorifics and respect forms; - grammar explanations; - anything shown to children.

Thresholds are starting points; tune them so that the **measured** error rate of Auto-Beta items stays under the tier target in §6.

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4. Pedagogical design

- **Outcomes:** CEFR can-do statements (A1–B1 first), mapped to ACTFL levels for US diaspora and schools. Each unit targets 2–4 can-dos ("I can greet an elder appropriately in the morning").
- **Vocabulary:** frequency-ranked from CC-BY corpora, adjusted for learner needs (family, food, market, church/mosque, transport, phone, money). Targets: **A1 about 600–800 lexemes, A2 about 1,500, B1 about 2,500–3,000** cumulative.
- **Spiral syllabus:** every pattern and high-frequency lexeme reappears at least 6 times across units in new contexts (input → controlled practice → speaking → review). FSRS schedules review.
- **Pattern Library:** one entry per real grammatical pattern per language, e.g. Yoruba preverbal *ń* (progressive), *ti* (perfect), *máa* (future), with examples, contrasts, common learner errors and citations. Swahili tense markers -na-/-li-/-ta-/-me-; Zulu subject concords by class.
- **Tone and sound strand:** perception and production drills from the lexicon's minimal pairs, plus FSI tone drills as references (yo, ig, ha); clicks for xh/zu; vowel purity and stress for sw.
- **Culture strand:** short, cited culture cards (greetings, kinship, food, festivals). Proverbs only after linguist review.
- **Item schema:** the elevation plan §6.5 schema plus `provenance.sources[]`, `claims[{text, source_id, locator}]`, `confidence`, `risk_tags[]`, `label` (beta | expert | native) and `audio.kind` (tts | human).

How items feed every feature (one content graph, many surfaces):

Content object	Feeds
Lexeme (lemma, tone, IPA, senses, audio, image, dialect variants)	Living Dictionary and Word Card; Point & Say object labels; Words & Review (FSRS); Tone Gym minimal pairs; games' word pools (Word Match, Memory Map, Talking Drum, Tone Forge); Lens OCR lookup

Content object	Feeds
Pattern	Grammar hub; Polie's grammar mode (answers only from patterns); Grammar Detective, Liar Liar; lesson explanations
Dialogue and scene	Lessons; Scenario Theatre; Market Haggle and Greeting Gauntlet scripts; Shadowing clips; Conversation Bridge "learn this phrase"
Exercise items (IRT-calibrated)	Lessons, Lesson Recap, checkpoints, placement test, Review Rush, Daily Challenges
Culture card (cited)	Culture shelf, Magazine, Heritage, Elders' Blessings, Proverb Weaver, **persona dossiers** (Historical Personas answer only from cited cards)
Can-do map	Path, Passport evidence, Progress, teacher dashboard

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5. Interim human quality without native authors

Mechanism	What it covers	Cost driver
Paid linguist reviewers (academics or postgrads in African-language linguistics; universities listed in elevation plan §6.4)	100% of high-risk items; 100% of the A1–A2 core lexicon (≈1,500 headwords, fast to check); a **random 10–15% sample** of Auto-Beta items	Per item, about 60–150 items/hour depending on type
Two-reviewer agreement on a calibration set (≈300 items per language)	Measures reviewer reliability (Cohen's κ) and calibrates the confidence score	One-off
A/B accuracy sampling	Each week a stratified random sample of live items is blind-reviewed; the publicly reported error rate per language and label comes from this	Ongoing, ≈200 items per language per month
Learner error flags	"Report a mistake" on every item and Word Card (category: wrong meaning, tone/spelling, audio, unnatural, offensive). Three or more flags, or one flag from a trusted user, auto-hides the item to "Under review"	Free; triage time
Item statistics	IRT difficulty and discrimination; abnormal patterns (e.g., experts failing an item) signal errors	Free
Community "suggest a fix"	Heritage learners propose corrections that feed the Verify queue (elevation plan §18, Village → Contribute)	Free; later native validators adjudicate

In-app labels (honest by design): - **Native-verified** (green): approved by ≥2 native validators. - **Expert-reviewed** (blue): approved by a linguist. - **Beta** (amber): passed all machine checks and sampling, but no human has checked this specific item. Copy: "AI-assisted, machine-checked. Spotted a problem? Tell us." - **Synthetic voice** tag on all TTS audio.

Rules: - Beta items never appear in **checkpoints, placement or credentials**. - Grammar and culture explanations are never Beta. - Children's mode shows Expert-reviewed or better only.

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6. Realistic accuracy per tier, and where AI-only is too risky

Measure, don't assert: "error" means a learner-visible mistake that changes meaning, misspells, mis-marks tone or is clearly unnatural, as judged in the blind sample. The figures below are **targets to verify in the pilot**, not measured results.

Tier	Languages	Why	Beta error target	Expert-reviewed target
A (lower risk)	Swahili	Large corpora, strong MT and LLM quality, no lexical tone, licensed dictionaries	≤ 3% of items	≤ 1%
B (medium)	isiZulu, isiXhosa, Hausa, Nigerian Pidgin	Zulu/Xhosa: complex morphology but unwritten tone and a Zulu analyser. Hausa: tone and length unwritten in normal text. Pidgin: spelling variation and LLM anglicisation	≤ 5%	≤ 1.5%
C (high tone risk)	Yoruba, Igbo	Meaning-bearing written tone marks; restorers reach about 80% (Yoruba) and about 62% (Igbo) word or tone accuracy on their own	≤ 5%, achieved **only by restricting Beta to lexicon-anchored tokens**; anything else goes to review	≤ 1.5%
Waves 2–4	Varies	Fewer sources	**No Beta below L0** unless a lexicon source exists; L0 "Taste" must be Expert-reviewed	≤ 2%

Where AI-only is too risky, and the mitigation:

Risk	Why	Mitigation
Tone marks (yo, ig; Hausa pedagogical marks)	LLMs and restorers are unreliable; a wrong mark changes the word	Lexicon-anchored only; three-way agreement; linguist for every unmatched token; tone teaching uses sourced minimal pairs only
Idioms and proverbs	Plausible fakes are easy to generate	Only from cited collections; linguist review; "Expert-reviewed" minimum
Culture, history and religion claims	Hallucinated or reductive claims, and sensitivity	Claim-level citations to reputable sources; cultural reviewer; no Beta
Honorifics and register	Social harm (addressing an elder with ọ)	Respect-sensitive units are Expert-reviewed only
Pidgin authenticity	Models write English with Pidgin spelling	Corpus-anchored phrasing (UD-Naija, NSC); anglicisation detector; early native validation priority
Dialect mixing (Igbo, Yoruba)	Models blend dialects	Declared standard in the language kit; dialect tag per item; variants only from sourced entries
Children's content	Higher duty of care	Expert-reviewed minimum

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7. Audio without native recordings yet

Interim policy. Every item has audio. Interim audio is **synthetic and labelled**, except where licensed human audio already exists for that exact text. Native recordings replace TTS in order of exposure.

Language	Interim TTS (from AI stack §3.5)	Licensed human audio usable now	Gate before use
Yoruba	Intron TTS (by quote) → Spitch fallback; always send fully tone-marked text	FSI Yoruba tapes (PD, dated quality) for tone drills; Common Voice and FLEURS for their own sentences	(1) ASR round-trip: synthesise → Intron ASR → text match; (2) **F0 contour versus the lexicon tone pattern** (automated tone proxy); (3) linguist listen-check of 100 samples
Igbo	Intron → Spitch	FSI Igbo tapes (PD); Common Voice, FLEURS	Same; Igbo tone proxy is weaker, so the linguist sample rises to 200

Language	Interim TTS (from AI stack §3.5)	Licensed human audio usable now	Gate before use
Hausa	Intron → Spitch / ElevenLabs v3	FSI Hausa (PD); Common Voice, FLEURS	Same
Nigerian Pidgin	Intron → Spitch	n/a	ASR round-trip + linguist listen
Swahili	Blind-test Intron vs Google Chirp 3 HD `sw-KE` vs Azure `sw-KE`	Common Voice (1,065 h), FLEURS	ASR round-trip + listen
isiZulu	Azure `zu-ZA` (Thando/Themba)	FLEURS, WAXAL (check subset licence)	ASR round-trip (Intron/Lelapa) + listen; check clicks
isiXhosa	**No commercial voice found.** Option A: text plus FLEURS/WAXAL audio where sentences match. Option B (recommended): commission a custom voice now (2 actors, 5–10 h each, buy-out)	FLEURS, WAXAL	Listen-check; clicks

Replacement plan: 1. As soon as native validators are contracted, record the **top 2,000 most-exposed items** per language first (A1 lexicon and dialogues); this typically covers most listening time. 2. Then record by exposure rank. 3. Each replacement bumps the item version and removes the "Synthetic voice" tag. 4. Keep TTS for dynamic content (Polie, personas) permanently, labelled.

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8. Rebuilding the existing curriculum data

1. **Quarantine.** Move the three placeholder bundles and the duplicated expanded bundle out of the app build (data, not features; the curriculum screens stay and read the new packs). CI fails on any placeholder pattern.
2. **Salvage.** Extract every real phrase from `lingafriq_authentic_curriculum_a1_c2.json` (one copy; the other two identical copies are dropped) and from `game_content.json`. Each becomes a **seed candidate** that goes through the full pipeline: lexicon anchoring, tone check, cross-check. Survivors keep their history (`provenance.source: legacy`); failures are logged for the linguist.
3. **Grammar.** Delete the two generic sentences from all 1,546 lessons and replace them with Pattern Library entries per language, written from cited grammars (FSI + reference grammars). The claim linter bans the old strings.
4. **Tone claims.** Every legacy `tonal_note` is re-derived from the lexicon's tone pattern. Claims like "High tone on àárọ̀ marks respect" fail the citation linter and are replaced with correct, cited notes (e.g., respect is carried by the honorific pronoun *Ẹ*). `phonetic_guide` is regenerated as IPA from G2P plus the tone pattern, never copied.
5. **Dialogues.** Regenerate from can-do targets with the anti-repetition linter (no repeated lines; each turn advances the scene).
6. **Structure.** Rebuild units against the CEFR map. One signed content pack per language and level, with content-hash audio filenames and NFC text.
7. **UI strings and phrase packs.** Run `app_*.arb` and offline phrase packs through the same orthography linters and linguist review (e.g., "Ẹ káàbọ̀ sí LingAfriq").
8. **Report.** Produce a before/after dashboard: items salvaged versus rebuilt, error rate in the sample, and coverage per CEFR level.

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9. Timeline, throughput, tooling and costs

9.1 Curriculum Studio (in `lingafriq-admin`)

Build it on the existing admin (33 pages, React/Vite), with a new backend job service. Features: - **Source library and licence register:** upload, licence fields, allowed-use flags, attribution text. - **Lexicon editor:** sources side by side, tone pattern, G2P preview, minimal pairs, anchored/single-source status. - **Generation jobs:** per can-do or unit, with prompt templates versioned in git. - **Verification report per item:** every model's opinion, back-translation, linter results, citations with the source passage highlighted, confidence score. - **Review queues:** high-risk, sample, flagged; keyboard-first, 1-click approve/fix/reject, diff view, reviewer agreement stats. - **Label and publish:** semver content packs, signing, staged rollout, rollback. - **Quality dashboard:** sample error rate per language and label, flags, item statistics, audio coverage, TTS versus human. - **Payments export** for reviewers (items reviewed per person).

Effort: **10–14 engineer-weeks** for the MVP (2 engineers × 6 weeks), plus a pipeline service of about 4–6 weeks **[estimate]**.

9.2 Phases

Phase	When	Output
0. Foundations	Weeks 0–4	Licence register signed by counsel; licences requested (Oxford, IgboAPI, Zulu dictionary, PanLex permission, NaijaVoices waiver if needed); 7 language kits; linters; Studio MVP; quarantine and salvage of legacy data; reviewer panels contracted (2 linguists per language, part-time)
1. Stage-1 A1 Beta	Weeks 4–12	A1 lexicon (600–800) anchored; A1 units, Pattern Library A1, culture cards (Expert-reviewed); TTS audio passing the gates; Beta labels live; weekly blind sampling
2. Stage-1 A2 + promotion	Weeks 12–24	A2 units; A1 high-exposure items promoted to Expert-reviewed; native validators onboard (later) and start verifying A1
3. Stage-1 B1 + Wave 2 L0	Months 6–9	B1 units; Wave 2 (14 languages) at L0 "Taste" (Expert-reviewed) once their kits and lexicon sources exist
4. Scale	Months 9–24	Wave 2 to A1 Beta; Wave 3 L0; Living Archive kits; native recordings replace TTS by exposure

9.3 Throughput (per language, after setup) [planning estimates, to measure in Phase 1]

- **Machine pipeline:** 2,000–4,000 items generated and verified per week (not the bottleneck).
- **Human review:** one part-time linguist (20 h/week) clears about **1,000–1,500 items/week**, at roughly 60–150 items/hour across lexicon checks, samples and high-risk items.
- **Publishable:** about **800–1,500 items/week** per language. A1 (≈3,000–4,000 items, including exercises) takes about 3–4 weeks; A1–B1 (≈10,000–12,000 items) takes about 10–14 weeks of review per language. The seven run in parallel.

9.4 Costs [rough estimates; get quotes]

Item	Per language (A1–B1)	Stage 1 (7 langs)
LLM generation + cross-checks + back-translation (≈10–12k items × ~4 model passes; Gemini 3.1 Pro / Flash + GPT-5-class; AI stack pricing)	$500–1,500	$4k–10k
Linguist review (≈150–300 h at $20–40/h local academic rates)	$3k–12k	$21k–84k
Calibration and blind sampling (ongoing, ≈200 items/month)	$100–200/month	≈$1k/month
Interim TTS (≈1–2 M characters; Azure $16/M chars; Intron by quote)	<$100 + quote	small + Intron quote

Item	Per language (A1–B1)	Stage 1 (7 langs)
isiXhosa custom voice (2 actors, buy-out, training)	n/a	$10k–25k [quote]
Data licences: Oxford datasets, IgboAPI, Zulu dictionary, NaijaVoices waiver ($5k+/yr), optional HCS (€15k)	varies	**$15k–60k one-off/annual [quotes]**
Reference grammars and dictionaries (print/ebook purchases)	$200–600	$2k–4k
Curriculum Studio + pipeline engineering	n/a	14–20 engineer-weeks

Rule of thumb: Stage 1 to A1–B1 Expert-reviewed for about **$65k–195k** all-in content cost (review, LLM, licences, Xhosa voice, first-year sampling), plus engineering, versus several times that for native authoring of the same volume. Each later language costs about **$4k–15k** to A1–B1 once its licences exist, because tooling, linters and prompts are shared.

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10. Decisions for Tope

1. Approve the **three-label policy** (Beta / Expert-reviewed / Native-verified) and the rule that Beta never appears in checkpoints, placement, credentials or children's mode. This is an interim change to the elevation plan's "two native approvals before publishing" rule (§6.6) and the AI stack's "curriculum audio is human-recorded" principle (§3.5): Beta and synthetic audio are allowed **with labels** until native validators arrive.
2. Approve licence purchases: **Oxford datasets** (top priority), **IgboAPI**, a **Zulu dictionary** licence, and optionally the NaijaVoices waiver and HCS.
3. Commission the **isiXhosa custom voice** now, or ship Xhosa with matched corpus audio plus text first.
4. Contract **2 part-time linguists per Stage-1 language** (universities in elevation plan §6.4), with a consent and IP clause.
5. Approve the Curriculum Studio build in `lingafriq-admin`.

*Sources were verified on 17 Oct 2026 via the linked pages. Items marked **[verify]**, **[purchase]** or **[quote]** need confirmation before contracts or spend. Not legal advice.*